

Clopen ordinal towers and the ball fixed point property for $C(K, p)$

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Source Question

The source paper is Aviles–Japon–Lennard–Martinez–Cervantes–Stawski, *The ball fixed point property in spaces of continuous functions*, arXiv:2506.17995 [1]. It asks in final Question (i) whether there is an infinite compact space K and a non-isolated point $p \in K$ such that

$$C(K, p) = \{f \in C(K) : f(p) = 0\}$$

has the ball fixed point property BFPP.

remainder is a much more delicate set-theoretic question. One can construct such P -points under the Continuum Hypothesis [29], but there are other models of set theory in which $\mathbb{N}^*$ contains no P -points [35].

We finally conclude the paper by highlighting several questions that may arise after reading the article, which we believe could be of significant interest for future research in the area:

- i) Is it possible to find an infinite compact set K and a non-isolated point $p \in K$ such that $C(K, p) = \{f \in C(K) : f(p) = 0\}$ has the BFPP?
- ii) Is it possible to construct an explicit example of a fixed point free nonexpansive mapping from the unit ball of ℓ_∞/c_0 into itself (without turning to CH)?
- iii) Is it possible to exhibit an example of a fixed point free nonexpansive mapping from the unit ball of $C(K)$ into itself, whenever $C(K)$ fails to be order-complete? In this case, we would have a topological characterization of the BFPP for Banach spaces of continuous functions: $C(K)$ satisfies the BFPP if and only if K is extremally disconnected.

References

The source paper proves failure when p is a non- P -point, and it also constructs special P -point examples for which $C(K, p)$ fails BFPP. A previous lane-15 packet proved failure for ordinal compacta $[0, \kappa]$ at the endpoint. The present packet records a reusable well-ordered clopen-tower mechanism.

Definition

Definition 1. Let K be compact Hausdorff and $p \in K$. A clopen ordinal tower at p is a family $(U_\alpha)_{\alpha < \kappa}$, where κ is a nonzero limit ordinal, such that:

1. $U_0 = K$, and each U_α is a clopen neighborhood of p ;
2. $U_\beta \subset U_\alpha$ whenever $\alpha < \beta < \kappa$;
3. $U_\lambda = \bigcap_{\alpha < \lambda} U_\alpha$ for every nonzero limit ordinal $\lambda < \kappa$;
4. $\bigcap_{\alpha < \kappa} U_\alpha = \{p\}$;
5. each annulus $A_\alpha = U_\alpha \setminus U_{\alpha+1}$ is nonempty.

The continuity condition at limit stages implies that

$$K \setminus \{p\} = \bigsqcup_{\alpha < \kappa} A_\alpha.$$

Indeed, if $x \neq p$, the least $\beta < \kappa$ such that $x \notin U_\beta$ cannot be 0 or a limit ordinal, hence $\beta = \alpha + 1$ and $x \in A_\alpha$. Also, compactness implies that the tower is a neighborhood base at p : if V is an open neighborhood of p , then the closed sets $(K \setminus V) \cap U_\alpha$ cannot all be nonempty, since their total intersection is empty.

Idea of the Proof

Choose one point $x_\alpha \in A_\alpha$ in each clopen annulus. For $f \in C(K, p)$, define Tf to be constant on each annulus: put 1 on the first annulus, shift the sampled value $f(x_\alpha)$ to the next annulus, and at a limit annulus put the ordinal limsup of the earlier sampled values. The annuli are clopen, so the only continuity issue is at p . But if $f(p) = 0$, then f is small on a sufficiently deep tower neighborhood, and all shifted and limsup values are small from that point onward.

The same formula is nonexpansive because point evaluations are contractive and limsup is 1-Lipschitz with respect to the sup norm. A fixed point would have sampled value 1 on A_0 , then on every successor annulus by the shift, and on every limit annulus by the limsup rule. The sampled points converge to p , contradicting $f(p) = 0$.

Partial Result

Theorem 1. *Let K be a compact Hausdorff space, let $p \in K$, and suppose that p admits a clopen ordinal tower. Then $C(K, p)$ fails the ball fixed point property.*

Proof. Let $(U_\alpha)_{\alpha < \kappa}$ be a clopen ordinal tower at p , and write $A_\alpha = U_\alpha \setminus U_{\alpha+1}$. Choose $x_\alpha \in A_\alpha$ for every $\alpha < \kappa$. We work over the real scalar field, as in the source paper.

For a bounded scalar family $(a_\alpha)_{\alpha < \lambda}$ indexed by ordinals below a nonzero limit λ , write

$$\limsup_{\alpha \rightarrow \lambda} a_\alpha = \inf_{\gamma < \lambda} \sup_{\gamma < \alpha < \lambda} a_\alpha.$$

Let $X = C(K, p)$. For $f \in B_X$, define Tf by $(Tf)(p) = 0$ and, for $x \in A_\beta$,

$$(Tf)(x) = c_\beta(f),$$

where

$$c_\beta(f) = \begin{cases} 1, & \beta = 0, \\ f(x_\alpha), & \beta = \alpha + 1, \\ \limsup_{\alpha \rightarrow \beta} f(x_\alpha), & \beta < \kappa \text{ is a nonzero limit ordinal.} \end{cases}$$

First, Tf is continuous and belongs to B_X . It is constant on each clopen annulus A_β , and all constants lie in $[-1, 1]$. It remains only to check continuity at p . Let $\varepsilon > 0$. Since $f(p) = 0$ and the tower is a neighborhood base at p , there is $\eta < \kappa$ such that $|f| < \varepsilon$ on U_η . If $\beta > \eta$, then for successor $\beta = \alpha + 1$ we have $\alpha \geq \eta$, hence $|c_\beta(f)| = |f(x_\alpha)| < \varepsilon$. If $\beta > \eta$ is a limit ordinal, the tail values $f(x_\alpha)$, $\eta < \alpha < \beta$, all have absolute value $< \varepsilon$, so $|c_\beta(f)| \leq \varepsilon$. Hence $|Tf| < \varepsilon$ on a sufficiently deep U_δ , and Tf is continuous at p .

The map $T : B_X \rightarrow B_X$ is nonexpansive. Let $f, g \in B_X$ and set $d = \|f - g\|_\infty$. On A_0 , $Tf - Tg = 0$. On a successor annulus $A_{\alpha+1}$,

$$|c_{\alpha+1}(f) - c_{\alpha+1}(g)| = |f(x_\alpha) - g(x_\alpha)| \leq d.$$

For a nonzero limit $\lambda < \kappa$, the elementary inequality

$$\left| \limsup_{\alpha \rightarrow \lambda} f(x_\alpha) - \limsup_{\alpha \rightarrow \lambda} g(x_\alpha) \right| \leq \sup_{\alpha < \lambda} |f(x_\alpha) - g(x_\alpha)|$$

gives $|c_\lambda(f) - c_\lambda(g)| \leq d$. Therefore $\|Tf - Tg\|_\infty \leq d$.

Finally suppose that $f \in B_X$ is a fixed point of T . Since Tf is constant on A_α , we have $f(x_\alpha) = c_\alpha(f)$. Transfinite induction gives $f(x_\alpha) = 1$ for every $\alpha < \kappa$. The initial step is $f(x_0) = c_0(f) = 1$. If $f(x_\alpha) = 1$, then

$$f(x_{\alpha+1}) = c_{\alpha+1}(f) = f(x_\alpha) = 1.$$

If $\lambda < \kappa$ is a nonzero limit ordinal and the conclusion holds for all $\alpha < \lambda$, then

$$f(x_\lambda) = c_\lambda(f) = \limsup_{\alpha \rightarrow \lambda} f(x_\alpha) = 1.$$

But the net $(x_\alpha)_{\alpha < \kappa}$ converges to p , because for every $\eta < \kappa$ and every $\alpha \geq \eta$, $x_\alpha \in U_\eta$. Continuity of f at p therefore forces

$$1 = \lim_{\alpha \rightarrow \kappa} f(x_\alpha) = f(p) = 0,$$

a contradiction. Thus T has no fixed point, and $C(K, p)$ fails BFPP. $\square$

Scope and Limitations

Theorem 1 is a genuine P -point-side obstruction whenever a well-ordered clopen tower exists. It does not settle the source question in full, because a non-isolated P -point need not come equipped with such a tower or with a norm-one retract onto a tower model. The arbitrary non-isolated P -point case remains the hard core of the problem.

Novelty and Verification Notes

The local run indexes were searched for arXiv 2506.17995, BFPP, $C(K, p)$, P -points, and clopen tower language. The only relevant prior run materials were the lane-15 ordinal compact partial packet and the unfinished full-push attempt. This packet promotes the clopen-tower route from that attempt into a precise partial theorem.

Human review should focus on two points: the continuity of Tf at p , and the 1-Lipschitz estimate for the limsup operation at limit annuli.

References

- [1] A. Aviles, M. Japon, C. Lennard, G. Martinez-Cervantes, A. Stawski, *The ball fixed point property in spaces of continuous functions*, arXiv:2506.17995, 2025.